

BCI for Mobile Devices: Time–Frequency and Representation Learning Analysis of Mobile Gesture Tasks

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Abstract—Mobile devices are central to daily life, yet their interaction remains primarily limited to touch and voice, which may be impractical in restricted mobility scenarios. Brain–computer interfaces (BCIs) offer a promising alternative by enabling device control through neural activity. Although motor imagery (MI) BCIs have been extensively studied, MI signals for mobile-specific gestures remain relatively underexplored. This study addresses this gap by investigating the effectiveness of combining time–frequency transformations with deep learning for classifying MI of mobile gestures. Using the MI-BMPI dataset—comprising EEG recordings of participants imagining tapping and swiping, and among the few publicly available datasets of its kind—we applied Short Time Fourier Transform, Continuous Wavelet Transform, Stockwell Transform, and Hilbert–Huang Transform as inputs to architectures including ResNeSt-50d, HRNet-W18, ConvNeXt-Base, DeiT-Base, Swin Transformer-Tiny, and RegNetX-002. In the classification of tapping and swiping MI tasks, performance across different subjects ranged from 0.61 to 0.947 when averaged over five random seeds, while individual subject models achieved results between 0.708 and 0.975. These promising outcomes are consistent with other evaluation metrics. Overall, this work provides the first systematic evaluation of mobile gesture MI-BCIs, demonstrating that time–frequency representations coupled with deep learning enable robust EEG-based mobile interaction and laying the groundwork for the development of future neuroadaptive technologies.

Index Terms—electroencephalography (EEG), brain–computer interface (BCI), human–computer interaction (HCI), mobile gesture, mobile neurofeedback, deep learning.

I. INTRODUCTION

Mobile devices such as smartphones and tablets have become an integral part of daily life, yet interaction is still predominantly achieved through touch or voice. In scenarios where mobility is restricted or the user’s hands are occupied, these modalities may be insufficient. BCIs offer an innovative alternative by enabling control through brain signals, bypassing the neuromuscular system. EEG-based BCIs have been applied across diverse domains. In robotics, they enable real-time control of robotic arms [1]; in virtual reality, they support applications such as drug rehabilitation [2]; in aerial systems, hybrid EEG–EMG frameworks improve drone safety [3];

and in medicine, BCIs aid in rehabilitation, communication, epilepsy detection, and anesthetic depth assessment [4]. Emotion recognition has also emerged as a distinct application, where deep learning achieves high accuracy in classifying affective states [5]. Collectively, these studies illustrate the versatility of EEG-based BCIs and the role of AI in processing neural signals. In MI-based BCIs, users imagine movements without physically executing them, and the corresponding EEG patterns are decoded into commands. Such systems are non-invasive, require no muscular activity, and have become feasible with recent advances in portable EEG technology. While most prior work has focused on imagined hand, foot, or tongue movements for computer interaction, mobile-specific gestures (e.g., tapping, swiping) remain largely unexplored, despite their practical relevance.

Traditionally, EEG classification relied on preprocessing, handcrafted feature extraction (e.g., CSP), and machine learning classifiers such as SVM or LDA. While effective, these methods require domain expertise and suffer from limited generalizability. In contrast, deep learning (DL) integrates feature extraction and classification, enabling automatic learning of complex EEG patterns. Recent studies have also employed time–frequency transformations (e.g., STFT, CWT, Stockwell) to convert non-stationary EEG into spectro-temporal images, thereby improving the discriminative power of DL models [6]. Despite these advances, EEG-based BCIs face challenges, including low spatial resolution, high dimensionality, and strong inter-/intra-subject variability [7]. Designing more adaptive DL architectures remains a key direction. Transfer learning is also promising, though EEG-specific adaptations and benchmark datasets are still limited. Finally, issues such as model interpretability, signal variability, and real-time usability highlight the need for more reliable, user-friendly MI-BCI frameworks.

This study addresses the gaps in the literature on BCI development by evaluating time–frequency transformation methods combined with deep learning architectures for classifying motor imagery EEG signals of mobile gestures—specifically tapping and swiping. The results demonstrate encouraging

performance and highlight the potential of this approach as a step toward advancing BCIs for mobile interaction.

II. MATERIALS AND METHODS

A. Experimental Data: Motor Imagery Brain–Mobile Phone Interface Dataset (MI-BMPI)

The MI-BMPI dataset was developed to address the lack of motor imagery EEG datasets specifically designed for brain–mobile device interfaces. It contains two MI classes representing common mobile phone gestures:

- Class 1: Motor imagery of tapping on the screen of a mobile device.
- Class 2: Motor imagery of swiping down with the thumb on the screen of a mobile device.

Eight healthy volunteers (aged 25–35; two female, six male) participated in the data collection. Each subject completed two sessions on separate days. EEG recordings were acquired using a consumer-grade, wireless Emotiv EPOC Flex neuroheadset from 13 electrode locations — C3, Cz, C4, and their surrounding sites — with a sampling rate of 128 Hz. The experiments followed a cue-based paradigm adapted from the BCI Competition IV datasets. For analysis, only the 320-sample segments corresponding to the motor imagery period between $t = 6$ s and $t = 8.5$ s were used. Each session comprised 96–120 trials per participant, balanced across the two MI classes and presented in random order. Two of the participants (D03 and D05) were female, while the remaining six were male. Only participant D07 was left-handed; all others were right-handed. During dataset creation, all subjects followed standardized instructions and identical recording protocols. Data quality was ensured through artifact removal, exclusion of noisy trials, and uniform preprocessing, which minimized inter-subject variability and maintained dataset consistency. Detailed information regarding the dataset is provided in the original study [8] in which it was created, and the download links for the dataset are also accessible through that publication.

B. Signal Preprocessing

The hierarchical architecture of deep learning networks enables automatic feature extraction from raw or minimally preprocessed EEG data, reducing the need for manual feature engineering and extensive preprocessing [9], [10]. Traditional machine learning approaches, by contrast, depend heavily on such steps. In this study, only basic preprocessing procedures—such as artifact removal and filtering—were applied before feeding transformed signals into the deep learning models.

For this study, only the motor imagery segments between $t=6$ – 8.5 s were used. Preprocessing was performed to remove noise and artifacts while preserving MI-related information. A band-pass FIR filter with a Hamming window (8–30 Hz) was applied to retain alpha and beta components and enhance the signal-to-noise ratio. Artifacts caused by eye blinks, muscle activity, and environmental noise were removed using Independent Component Analysis (ICA) via EEGLAB’s [11]

pop_runica function with the extended infomax algorithm. Independent components were automatically classified using ICLabel [12], and those with a 90–100% probability of representing artifacts were discarded.

C. Transformation Methods for EEG Analysis

To investigate the time-varying spectral characteristics of EEG signals associated with motor imagery tasks, four time–frequency analysis methods were employed. All transformations were implemented in Python using relevant libraries. The outputs of all transformation methods were resized to 224×224 pixels to ensure compatibility with the deep learning models. An example EEG signal from the C3 channel of the fourth trial in participant D01’s first session, along with its time–frequency representations obtained using the transformation methods described above, is shown in Figure 1.

1) *Short-Time Fourier Transform (STFT)*: STFT was applied to compute spectrograms for EEG segments by dividing the signal into short, overlapping windows and computing the Fourier transform for each segment. The sampling rate was set to 128 Hz, and the frequency range of interest was 4–30 Hz. A window size of 64 samples, an FFT length of 128 points, and 90% window overlap were used. The Hann window function was applied to minimize spectral leakage. The magnitude-squared of the complex STFT coefficients was computed to obtain the power spectrum, which was then visualized in the time–frequency domain. An example EEG signal is shown in Figure 1 (a), and the spectrogram obtained by applying the STFT to this signal is presented in Figure 1 (b).

2) *Continuous Wavelet Transform (CWT)*: CWT was computed using the PyWavelets with the complex Morlet (“amor”) wavelet, focusing on the 4–30 Hz range. The magnitude-squared of the CWT coefficients was calculated to obtain the scalogram representation. An example EEG signal is shown in Figure 1 (a), while the scalogram obtained by applying the CWT to this signal is presented in Figure 1 (c).

3) *Stockwell Transform (ST)*: The ST was implemented using the *s_transform* library in Python. EEG segments from each channel were transformed within the 4–30 Hz frequency range. The absolute values of the complex coefficients were normalized using min–max scaling to the range [0, 1], then visualized using the “jet” colormap. An example EEG signal is shown in Figure 1 (a), whereas the spectrogram derived from applying the ST to this signal is displayed in Figure 1 (d).

4) *Hilbert–Huang Transform (HHT)*: For HHT analysis, empirical mode decomposition (EMD) was first applied using the PyEMD library to adaptively decompose EEG signals into intrinsic mode functions (IMFs) without predefined basis functions. Only the first six IMFs were retained for further analysis, as these typically capture the most relevant frequency components. The Hilbert transform was then applied to each IMF to compute instantaneous amplitude and frequency. Frequency bins were limited to the 4–30 Hz range, divided into 100 bands to focus on MI-related alpha and beta

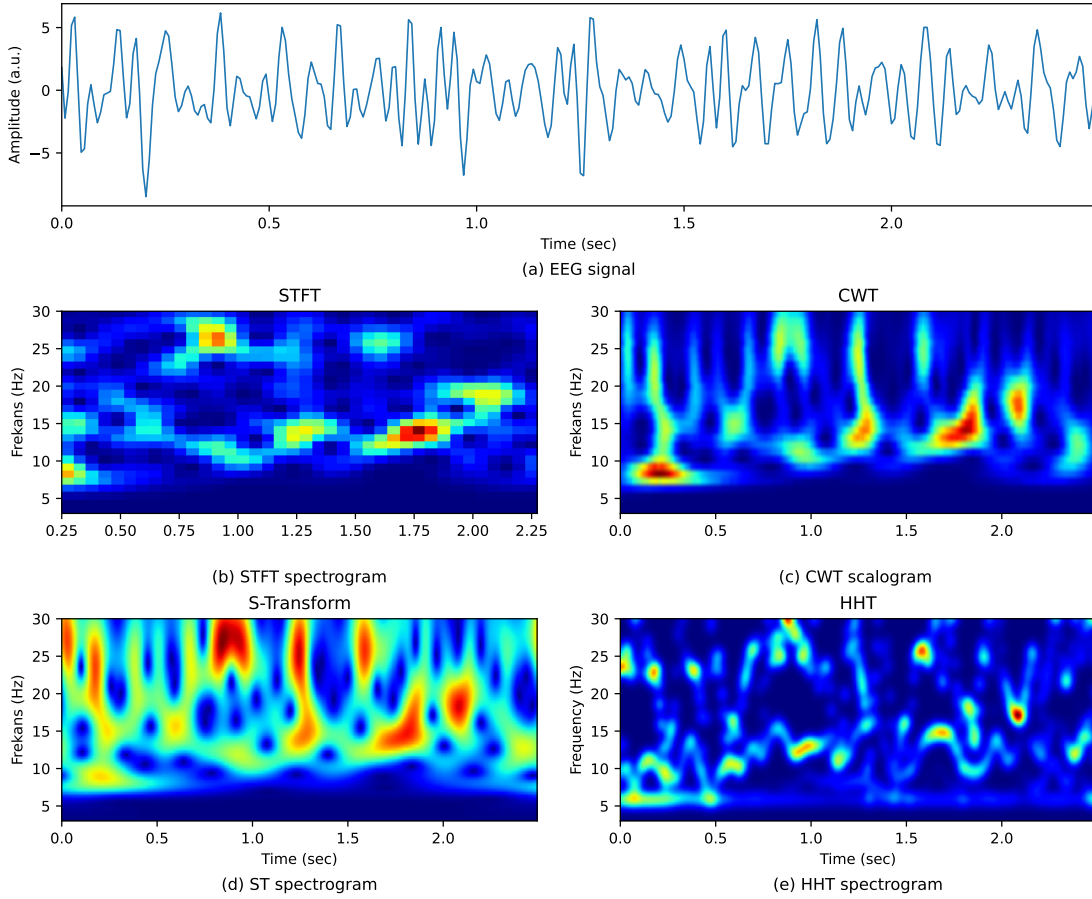


Fig. 1. Example EEG signal and representations obtained through different transformations.

rhythms. For each IMF, the amplitude values below 5% of the maximum were discarded to remove low-energy oscillations. An example EEG signal is shown in Figure 1 (a), whereas the spectrogram derived from applying the HHT to this signal is displayed in Figure 1 (e).

D. Deep Learning Models

1) *ResNeSt-50d*: ResNeSt-50d is an improved variant of the conventional ResNet-50 model, distinguished primarily by the integration of the Split-Attention mechanism, which significantly strengthens feature extraction capabilities [13]. In this mechanism, channels are divided into separate groups, allowing attention to be calculated independently for each group. The outputs are then dynamically combined, enabling more effective modeling of inter-feature map relationships. While the residual block structure of the original ResNet is preserved, the addition of channel grouping and intra-group attention substantially improves its performance. This design ensures that the network emphasizes the most informative channels, resulting in richer and more discriminative feature representations. By directing the flow of channel-level information more effectively, ResNeSt-50d enhances multi-scale representation power and achieves finer distinctions between visual elements.

2) *HRNet-W18*: The High-Resolution Network (HRNet-W18) represents a CNN architecture explicitly tailored for high-resolution representation learning [14]. In contrast to conventional CNN designs, which inherently diminish spatial granularity due to progressive downsampling operations, HRNet introduces a paradigm in which parallel convolutional branches at multiple resolutions are maintained throughout the network. These parallel streams are interconnected via continuous bidirectional information exchange, thereby enabling the preservation of both high- and low-resolution contextual cues. This architectural strategy effectively mitigates the spatial information loss typically observed in sequential convolutional hierarchies such as ResNet. By sustaining high-resolution feature maps, HRNet facilitates the extraction of fine-grained structural characteristics and enhances the fidelity of spatial detail representation.

3) *ConvNeXt-Base*: The ConvNeXt [15] was introduced as a modernized convolutional framework that reinterprets conventional CNNs through the integration of design principles inspired by transformer-based models. While preserving the sequential convolutional paradigm inherent to traditional CNNs, ConvNeXt strategically incorporates architectural refinements to bridge the performance gap with transformer networks. Specifically, it adopts enlarged kernel sizes to

capture broader contextual dependencies and employs depth-wise convolutions to counterbalance the associated computational overhead. Furthermore, the architecture leverages advanced normalization strategies, which contribute to stable optimization dynamics and improved scalability during training, thereby facilitating more efficient and robust feature extraction. The ConvNeXt-Base configuration demonstrates superior parameter efficiency and consistently achieves higher predictive accuracy than conventional convolutional networks.

4) *DeiT-Base*: The DeiT-Base (Data-efficient Image Transformer – Base) architecture was proposed as a streamlined variant of the Vision Transformer (ViT) that seeks to mitigate the substantial data and computational demands typically associated with pure attention-based models [16]. A key contribution of this framework is the introduction of a distillation token within the transformer pipeline, which enables the adoption of a teacher–student learning paradigm. In this configuration, a high-capacity CNN (e.g., RegNetY-16GF) functions as the teacher, transferring inductive biases to the student transformer through attention-based distillation. This design substantially improves data efficiency, accelerates convergence, and enhances predictive performance compared to standard ViT models.

5) *Swin Transformer-Tiny*: The Swin Transformer-Tiny (Swin-T) architecture was introduced as a hierarchical ViT to overcome the scalability and locality limitations of vanilla ViT models [17]. Swin adopts a hierarchical representation by progressively merging image patches, thereby reducing sequence length and enabling efficient modeling of high-resolution inputs. A key innovation is the shifted window mechanism, which confines self-attention to local non-overlapping windows while allowing cross-window communication through periodic shifts. This design yields linear computational complexity with respect to image size while still capturing long-range dependencies. The Swin-Tiny configuration represents the smallest variant in the Swin Transformer family, comprising approximately 28.3M parameters, and is particularly suitable for tasks with constrained computational resources.

6) *RegNetX-002*: The RegNetX-002 model belongs to the RegNet family of CNNs, which was proposed to systematically explore network design spaces through simple and regular structural principles [18]. RegNet introduces a parameterizable design space, where architectural dimensions are governed by quantized functions. This formulation yields a highly scalable family of models ranging from lightweight to large-scale architectures, while maintaining consistent design principles. The RegNetX series represents a simplified variant that removes squeeze-and-excitation blocks, with a primary emphasis on convolutional efficiency. Among these, the RegNetX-002 configuration is one of the smallest, comprising approximately 2.7M parameters and 16 million FLOPs, which makes it particularly well-suited for deployment in resource-constrained environments. This balance of simplicity, scalability, and efficiency positions RegNetX-002 as a representative baseline for evaluating lightweight convolutional networks.

7) *Overall Comparison of Architectures*: In terms of architectural characteristics and computational demands, the employed models exhibit notable differences. ConvNeXt-Base, influenced by transformer design principles, demonstrates the highest computational burden due to its enlarged kernels, depth-wise convolutions, and advanced normalization schemes, yet this complexity yields state-of-the-art accuracy across diverse vision tasks. ResNeSt-50d enhances the conventional ResNet framework by incorporating split-attention mechanisms, which moderately increase architectural depth and parameter size, positioning its complexity between HRNet-W18 and ConvNeXt-Base. In contrast, HRNet-W18 adopts a multi-resolution parallel branch design with continuous information exchange, maintaining high-resolution representations at a relatively lower computational cost while excelling in tasks requiring fine-grained spatial detail. DeiT-Base, as a data-efficient ViT, introduces self-attention mechanisms that scale linearly with input sequence length, leading to higher complexity than convolution-based backbones but optimized training strategies allow effective use of moderate resources. Swin-Tiny reduces transformer complexity through hierarchical shifting windows, achieving a favorable trade-off between efficiency and representational power, making it suitable for dense vision tasks with limited computational budgets. Finally, RegNetX-002 emphasizes parameter efficiency and scalability through a simple, regularized design, resulting in the lowest computational complexity among the employed models, albeit with comparatively reduced expressive capacity.

Since the dataset used in this study consists of motor imagery data representing mobile device gestures, it is important to evaluate the proposed models in terms of computational complexity and their applicability in real-world scenarios. In this context, ConvNeXt-Base and ResNeSt-50d require tens of millions of parameters and multiple GFLOPs, whereas HRNet-W18, DeiT-Base, and Swin-Tiny demand moderate computational resources. RegNetX-002, with approximately 2.7M parameters and 16M FLOPs, stands out as the most lightweight model and thus represents the most suitable option for deployment on mobile devices and in real-world settings.

III. EXPERIMENTS, RESULTS & DISCUSSION

In this study, we conducted an extensive experimental framework to evaluate the performance of multiple state-of-the-art deep learning architectures on subject-specific EEG classification tasks using the MI-BMPI dataset. All recordings were preprocessed, filtered, and transformed into image-based representations using different time–frequency analysis techniques (e.g., CWT, HHT, STFT, and ST). We focused on motor cortex channels by selecting C3, Cz, and C4 according to their standard positions in the international 10–20 system, which are well known to overlie motor-related cortical areas. In experiments, for each subject, EEG recordings from the first and second sessions were merged to form a single dataset, ensuring that all available trials were utilized. The combined data was then randomly split into training (80%), validation

(10%), and test (10%) subsets to maintain class balance and provide a fair evaluation protocol.

A comprehensive hyperparameter search was performed to investigate the influence of architectural choices and training settings on classification performance. The experimental design covered six representative pre-trained models in the timm open-source library—ResNeSt-50d (resnest50d), HRNet-W18 (hrnet_w18), ConvNeXt-Base (convnext_base), DeiT-Base (deit_base_patch16_224), Swin Transformer-Tiny (swin_tiny_patch4_window7_224), and RegNetX-002 (regnetx_002)—spanning both convolutional and transformer-based feature extractors. Three optimizers (Adam, SGD, and RMSprop) were combined with three learning rates (1e-5, 1e-4, 1e-3) and two batch sizes (16 and 32), resulting in 108 unique configurations per seed. Performance evaluation was based on multiple metrics, including best validation accuracy, mean accuracy, recall, precision, specificity, F1-score and Kappa. This exhaustive search enabled the identification of the optimal combination of transformation method, EEG channel, model architecture, and hyperparameters for each subject. To ensure robustness and reproducibility, each configuration was evaluated across five independent random seeds. The final performance metrics were calculated by averaging the results obtained from these five runs. The best-performing configurations for subjects D01–D08, according to mean accuracy, are summarized in Table I.

As shown in Table I, the results demonstrate considerable variability across subjects, highlighting the subject-specific nature of EEG-based MI task classification. Overall, the ST and HHT provided the most consistent performance, particularly when applied to the C3 and C4. In contrast, STFT and CWT exhibited relatively lower performance, with mean accuracies around 0.6 for some subjects. Among the evaluated architectures, ConvNeXt-Base achieved the strongest performance in multiple cases (D03, D06, D07), with D07 yielding the highest overall mean accuracy of 0.947. This result illustrates the potential of modern convolutional architectures to capture discriminative EEG patterns, especially when trained on subject-specific data. In contrast, transformer-based models such as DeiT-Base demonstrated competitive results (e.g., D02 with 0.625 accuracy), though generally underperformed compared to ConvNeXt across most subjects. HRNet-W18 showed stable but moderate performance, while RegNetX-002 achieved promising results for D08, reaching 0.736 accuracy on the C4 channel with HHT features. Swin Transformer-Tiny, however, did not yield the best average outcomes for any subject. Beyond accuracy, the remaining performance metrics were largely consistent with the accuracy values, indicating balanced classification behavior across classes. Importantly, the Kappa coefficient followed a similar trend: subjects with higher mean accuracy (e.g., D07 with 0.947) also exhibited substantially higher Kappa values (0.892), whereas lower-accuracy subjects achieved correspondingly modest Kappa scores (e.g., D05 with 0.627 accuracy and 0.218 Kappa). This consistency across metrics strengthens the validity of the reported results and confirms the discriminative capacity of the

selected pipelines. The variation in optimal channels across subjects highlights the neurophysiological relevance of motor cortex regions and the need for personalized model selection. These findings show that no single method dominates; optimal performance arises from tailoring the transformation, channel choice, and model architecture to each individual.

In addition to the averaged results across seeds, we also examined the maximum performance achieved for each subject by selecting the best outcome among all seed runs (Table II). This complementary analysis illustrates the upper-bound potential of the proposed methods and demonstrates that, under favorable initialization conditions, substantially higher accuracies can be obtained. The results indicate that the ST yielded the most consistent high performance across subjects, with RegNetX-002, ConvNeXt-Base, and DeiT-Base emerging as the best-performing architectures in different cases. For example, subject D07 achieved an exceptional best accuracy of 0.975 with ConvNeXt-Base on the C3 channel, highlighting the capacity of modern convolutional networks to capture discriminative MI features. Similarly, RegNetX-002 achieved robust results for D01 and D05, while DeiT-Base surpassed other models for D04 with both strong accuracy (0.762) and high precision (0.853). Interestingly, the maximum-performance evaluation shows that some subjects (e.g., D06, D08) achieve much higher accuracies than suggested by averaged-seed experiments, underscoring the variability from random initialization and the benefits of multiple runs. While average-seed results provide a stable estimate, further improvements may be achieved through ensemble strategies or careful re-initialization. Beyond accuracy, other metrics followed similar trends, and Kappa values also increased accordingly, with top cases (e.g., D07 with ConvNeXt-Base) showing high agreement well beyond chance.

Brain-computer interface (BCI) research has increasingly expanded toward mobile applications, and such approaches can indeed be adapted for mobile device interaction. One of the earliest efforts was the study [19], which proposed an assistive framework called Neuro-phone to operate smartphones using EEG signals for people with disabilities. The framework relied on Discrete Fourier Transform and a Hidden Markov Model classifier applied to EEG signals representing nine mental commands from eight participants, achieving an average accuracy of 68.69%. The results demonstrated the feasibility of EEG-based smartphone operation and highlighted the potential of such approaches for assistive and healthcare applications. Building on this line of work, the NeuroPhone study [20] further explored direct brain-mobile device interaction using low-cost wireless EEG headsets. By leveraging neural responses such as P300 potentials and wink detection, the system enabled hands-free and silent smartphone control. The study reported promising results for wink classification, achieving precision, recall, and accuracy of 92.35%, 99.39%, and 95.58% in sitting-relaxed conditions, and maintaining robust performance during walking with 86.15%, 96.70%, and 92.58%, respectively. Another related effort is the YY

TABLE I
BEST RESULTS PER SUBJECT BASED ON AVERAGE OF SEEDS

Subject	Transformation	Model	Channel	Mean Acc	Mean Rec	Mean Pre	Mean Spe	Mean FM	Kappa
D01	HHT	ResNeSt-50d	Cz	0.619	0.625	0.616	0.625	0.620	0.229
D02	STFT	DeiT-Base	C3	0.625	0.622	0.626	0.622	0.624	0.239
D03	ST	ConvNeXt-Base	C4	0.638	0.638	0.661	0.638	0.649	0.273
D04	CWT	HRNet-W18	C3	0.610	0.622	0.621	0.622	0.622	0.237
D05	STFT	HRNet-W18	C4	0.627	0.613	0.675	0.613	0.639	0.218
D06	ST	ConvNeXt-Base	C3	0.795	0.703	0.737	0.703	0.719	0.419
D07	ST	ConvNeXt-Base	C3	0.947	0.948	0.947	0.948	0.947	0.892
D08	HHT	RegNetX-002	C4	0.736	0.726	0.749	0.726	0.737	0.450

TABLE II
BEST RESULTS PER SUBJECT AMONG DIFFERENT SEEDS

Subject	Transformation	Model	Channel	Acc	Rec	Pre	Spe	FM	Kappa
D01	ST	RegNetX-002	C4	0.762	0.683	0.706	0.683	0.695	0.386
D02	ST	ConvNeXt-Base	C3	0.708	0.707	0.703	0.707	0.705	0.408
D03	ST	ResNeSt-50d	C3	0.714	0.718	0.722	0.718	0.720	0.432
D04	ST	DeiT-Base	C3	0.762	0.722	0.853	0.722	0.782	0.478
D05	ST	RegNetX-002	C4	0.773	0.641	0.667	0.641	0.654	0.304
D06	ST	ConvNeXt-Base	C3	0.837	0.771	0.789	0.771	0.780	0.559
D07	ST	ConvNeXt-Base	C3	0.975	0.975	0.975	0.975	0.975	0.949
D08	CWT	RegNetX-002	C4	0.818	0.830	0.808	0.830	0.819	0.627

study [21], which introduced a prototype BCI solution based on the NeuroSky MindWave Mobile device. In this system, EEG signals and eye blinks were analyzed through custom software tools and mapped to actions on smartphones or tablets. Experiments with different configurations, including single or double blinks and varying focus levels, showed that users could reliably and touchlessly control mobile applications, further demonstrating the promise of EEG-based mobile interaction.

Studies focusing on performing mobile-specific gestures through motor imagery signals remain quite limited. One such effort is the work by [8], which introduced the MI-BMPI dataset with two mobile-related MI gestures and proposed voting ensemble methods based on frequency bands and time segments. The experiments reported classification accuracies between 78.4% and 86.25% across subjects for the two-class mobile gesture motor imagery tasks. In contrast, the present study extends this line of research by systematically evaluating time–frequency transformations in combination with state-of-the-art deep learning architectures. The results demonstrate considerably higher performance in several cases—for instance, ConvNeXt-Base achieved a mean accuracy of 0.947 and a maximum accuracy of 0.975 for subject D07—while also providing a more comprehensive evaluation through additional metrics such as recall, precision, specificity, and F1-score. These findings not only confirm the feasibility of MI-based mobile interaction but also highlight the advantages of integrating modern convolutional architectures and time–frequency representations for robust EEG classification.

IV. CONCLUSION AND FUTURE WORKS

This study systematically explored motor imagery–based BCIs for mobile gestures, combining time–frequency transformations with deep learning models. Results showed subject-specific variability, with ConvNeXt and RegNetX achieving the most consistent performance on motor cortex channels. These findings confirm the feasibility of EEG-based mobile interaction and highlight the potential of adaptive strategies for future BCI systems. In addition, the comparison of averaged and maximum-performance outcomes revealed that initialization effects play a critical role, suggesting that ensemble learning or re-initialization strategies could further enhance reliability. The consistent alignment of accuracy with other evaluation metrics, particularly the Kappa coefficient, strengthens the robustness of the proposed framework. Importantly, the variation in optimal EEG channels across subjects highlights the need for personalized model selection in practical BCI applications.

Building on these results, future research directions can be outlined. Looking forward, increasing the number of samples in datasets via data augmentation is an important research direction. In addition, new transformation methods should be proposed to provide more informative inputs for deep neural networks, and novel architectures specifically tailored for EEG signals should be developed. To better understand the classification mechanisms of EEG signals, future work should also incorporate explainable AI approaches, enabling greater interpretability and transparency in BCI systems. Future studies should also develop software solutions that integrate time–frequency images with CNNs, which could serve not only brain–computer interfaces but also critical

applications such as diagnosing and treating neurological diseases. Furthermore, the adoption of MLOps practices can optimize model development, versioning, automated training, deployment, and monitoring processes for EEG signals, thereby enabling the creation of more reliable, scalable, and reproducible AI models for both BCI and healthcare domains. As part of our future project, we also aim to implement the model obtained in this study within an MLOps pipeline, ensuring practical deployment and continuous improvement. In addition, in future, integrating the trained model into a mobile app prototype with real-time feedback will be highly beneficial, enabling closed-loop neuroadaptive control.

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